

Retail Sales Analysis & Dashboard Project

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Role: Data Analyst (Project)

Tools Used: Python, SQL, Excel

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Project Overview

This project focuses on analyzing retail sales data using a complete data pipeline involving **Python (Pandas), SQL, and Excel**. The goal is to transform raw data into meaningful insights and present them through an interactive dashboard.

Project Workflow

1. Data Cleaning (Python - Pandas)

- Imported raw Superstore dataset
 - Handled missing values and duplicates
 - Converted date columns to proper datetime format
 - Cleaned inconsistent data entries
 - Exported cleaned dataset for further analysis
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2. Data Analysis (SQL)

- Performed queries to extract insights:
 - Total Sales & Profit
 - Category-wise performance
 - Region-wise analysis
 - Monthly sales trends
 - Top 5 products by sales
 - Used aggregation functions (SUM, COUNT, GROUP BY)
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3. Data Visualization (Excel Dashboard)

- Built an interactive dashboard using:
 - Pivot Tables
 - Pivot Charts
 - Slicers (Category, Region, Segment, Ship Mode)
 - KPI Cards
 - Conditional Formatting (Trend Indicators ▲ ▼)
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Dashboard Features

KPI Metrics

- Total Sales
 - Total Profit
 - Total Orders
 - Average Order Value
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Sales Distribution

- Category-wise breakdown (Pie Chart)
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Top Products Analysis

- Top 5 products based on sales (Bar Chart)
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Monthly Performance

- Sales vs Profit trend (Line Chart)
 - Trend indicators (▲ ▼) to show increase/decrease
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Interactive Filters

- Category

- Region
- Segment
- Ship Mode

👉 Dashboard dynamically updates based on slicer selection

Key Insights

- Technology category contributes the highest sales
 - Profit trends fluctuate across months
 - Some months show declining profit despite stable sales
 - Top products contribute significantly to revenue
 - Regional performance varies significantly
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Tools & Technologies Used

- Python (Pandas)
 - SQL
 - Microsoft Excel
 - Pivot Tables & Charts
 - Data Cleaning & Transformation
 - Data Visualization
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Conclusion

This project demonstrates the complete data analysis lifecycle—from raw data cleaning to insight generation and visualization. The dashboard enables quick decision-making using interactive and visual insights.

Future Enhancements

- Add Profit Margin analysis
- Year-over-Year comparison

- Power BI Dashboard version
- Automated data refresh

Dashboard Preview

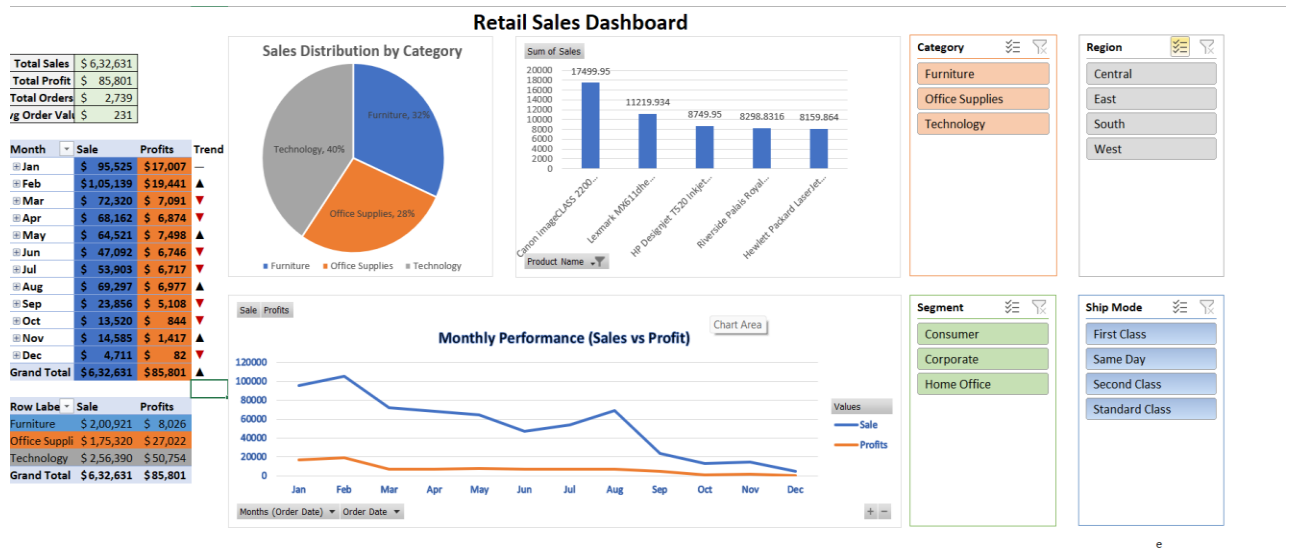


Figure 1: Interactive Retail Sales Dashboard

Project Structure

Project_2_Superstore_Analysis/

data/ → Raw & Cleaned dataset

python/ → Data cleaning scripts (Pandas)

sql/ → SQL queries for analysis

dashboard/ → Excel dashboard file

reports/ → Project report